V300 ZEFIRO

A New Sense of Speed

# Interior Doors-VF.CONCEPT SFRS

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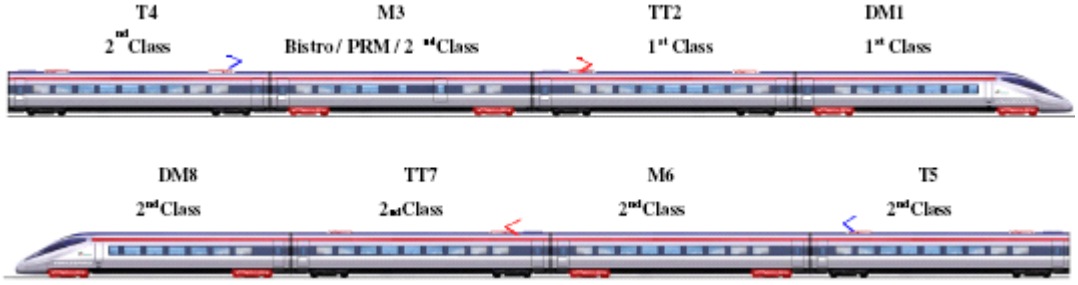
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# 1 Introduction

This document is a function specification for the control of the Interior doors system for V300ZEFIRO for customer Trenitalia.

## 1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:




- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

## 1.2 Definitions and abbreviations

| | |
|---|---|
| CD | Circuit Diagram |
| EVENT | Diagnostics that will be stored in the Event database within TCMS SW. |
| HW HMI | Hardware Human Machine Interface, e.g. buttons, switches and indication lamps. |
| IDU | Interface Display Unit |
| TCMS | Train Control and Management System |
| FDS | Fire Detection System |
| DCU | Door Control Unit |
| SW | Software |

# 2 System Overview

The interior doors system includes automatic saloon doors and gangway doors.

***Automatic saloon doors***

It is a single leaf sliding door electrically driven. There are two kinds of saloon door (STD saloon door and HK saloon door) differing only for the opening free width (700mm Vs 900mm)

For each saloon door a dedicated Door Control Unit (DCU) is foreseen. The DCU manages all the door functionalities, particularly:

- opening command (by presence sensors and by push buttons on both side of the door)
- closing command (by internal timer)
- passenger protection system (jamming protection)
- diagnostic

Among the STD saloon doors, the 2 doors in DM1 coach confining the galley area are called Galley doors and differ from the STD saloon doors for the fact that presence sensors are installed only on the passenger saloon and vestibule side to prevent that the presence of galley area staff result in a permanent door open command. Moreover a dedicated function allows the synchronization in the opening of the 2 doors to improve passenger comfort.

Quantities and distribution of the internal doors on the train are shown in the *Figure 1 – Internal doors identification/layout.*

Each saloon door is connected to the TCMS by means of two hardwired lines (for diagnostic purpose only).

***Gangway doors***
It is a double leaf sliding door manually operated provided with force closing device (spring). During normal operation the gangway door is locked in its fully open position. In case of fire, the locking device is released (by the fire detection system) and the door closes automatically (by spring). The closing position is identified by a suitable switch and sent to the TCMS for diagnostic purpose.
When closed, the door is operable by hand.

The gangway door of the M3 coach on the bistro side (indicated as GWDoor_R_M3 in picture 1) differs from the others because it is a normally closed double leaf sliding door electrically driven provided with force closing device (spring). The dedicated Door Control Unit (DCU) manages all the door functionalities (as saloon doors), particularly:

- opening command (by push buttons on both side of the door)
- closing command (by internal timer)
- passenger protection system (jamming protection)
- diagnostic
- opening inhibition function (fire protection)

In case of fire the door will close (if open) and remain closed by cutting the power supply to the control unit. When closed, the door is operable by hand.
This door can be locked in its fully open position in case of any failure or for maintenance. If locked in open position, the door has the same behaviour of the standard gangway doors.

## 2.1 System Configuration

The internal doors are identified according to the following rule:

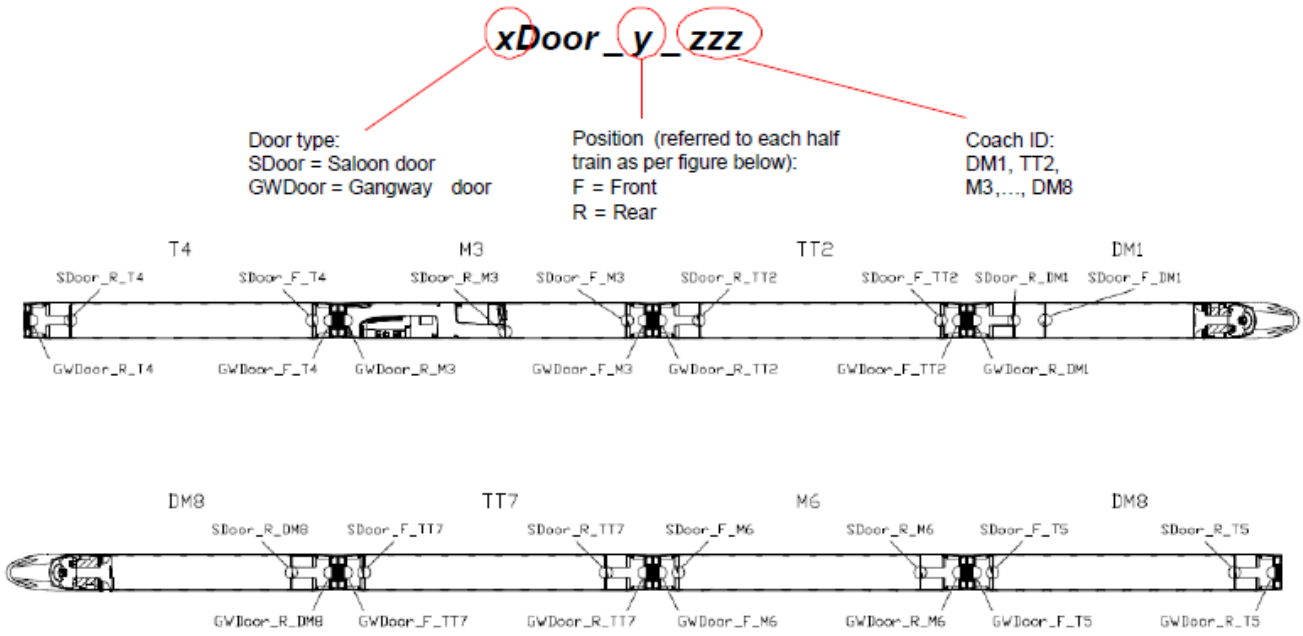



*Figure 1 – Internal doors identification/layout*

# 3 Functional requirements

## 3.1 Gangway Doors

### 3.1.1 Gangway Doors - Fire mode

#### 3.1.1.1 Normally open gangway doors

Following sub sections are applicable to all the standard gangway doors and to the GWDoor_R_M3 if it has been locked in open position.

##### 3.1.1.1.1 Function Design

The gangway doors closing commands are realized by means of hardwired lines defined in the Fire Detection System (FDS) SFRS. The logics, in terms of which door has to be closed in case of fire, are handled by the fire detection system. Each gangway door will close as soon as the relevant closing command is energized. In parallel to the FDS command a thermal fuse is able to close the door if the temperature near the gangway door is greater than a predefined threshold.

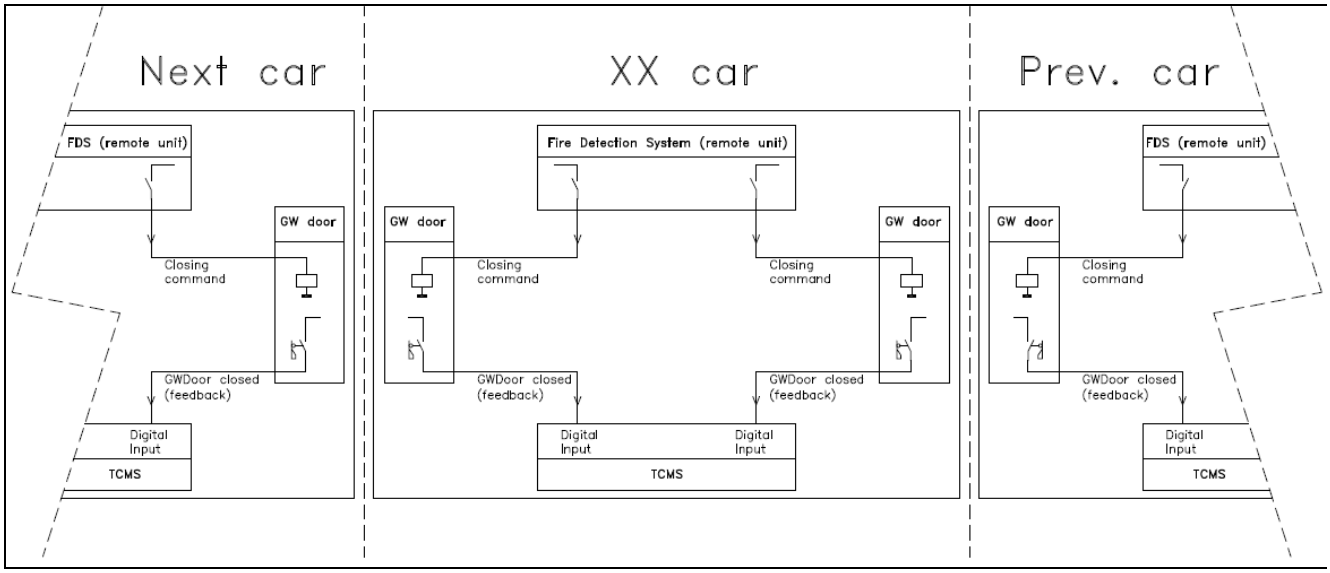



Figure 2 - Gangway door closing command block diagram (coach level)

##### 3.1.1.1.2 Normal condition

In case of fire, when the gangway door receives the relevant command from the FDS, the locking device (electrical motor) is released and the closing device (spring) closes the door leafs.
If the temperature near to the gangway door mechanism is above a preset value, a thermal fuse melts and releases the locking device starting the closing sequence, independently by the FDS status.

##### 3.1.1.1.3 Failure cases

a) Loss of function of the locking device (electrical motor). Loss of a redundancy (the thermal fuse is still able to close the door).
b) Thermal fuse doesn't melt but overtemperature is present. Loss of a redundancy (the FDS is still able to close the door).

c) Thermal fuse melts but no overtemperature is present. Undue closing of the door (it can be opened manually)
d) Loss of function of the closing device (spring).The gangway doesn't close.

#### 3.1.1.1.4 Failure detection

a) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door
b) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door
c) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door
d) Failure condition detected by the TCMS comparing the FDS command with the current position of the gangway door

#### 3.1.1.1.5 Failure handling

a) Driver has to act according to operational manual
b) Driver has to act according to operational manual
c) Melted thermal fuse has to be changed during maintenance.
d) Driver has to act according to operational manual

### 3.1.1.2 Normally closed gangway door

Following sub sections are applicable to the GWDoor_R_M3 in its normal functioning, i.e. when the door is not locked.

#### 3.1.1.2.1 Function Design

The closing command is realized by means of an hardwired signal defined in the Fire Detection System (FDS) SFRS. The need of closing (if open) or keeping this door closed (if already closed) in case of fire is handled by the fire detection system. As soon as the relevant command from the fire detection system is energized, the power supply will be cut to the DCU by means of hard wired circuits in such a way that the DCU can not command any door movement. As a redundancy, the DCU SW will inhibit all the undue door opening command.

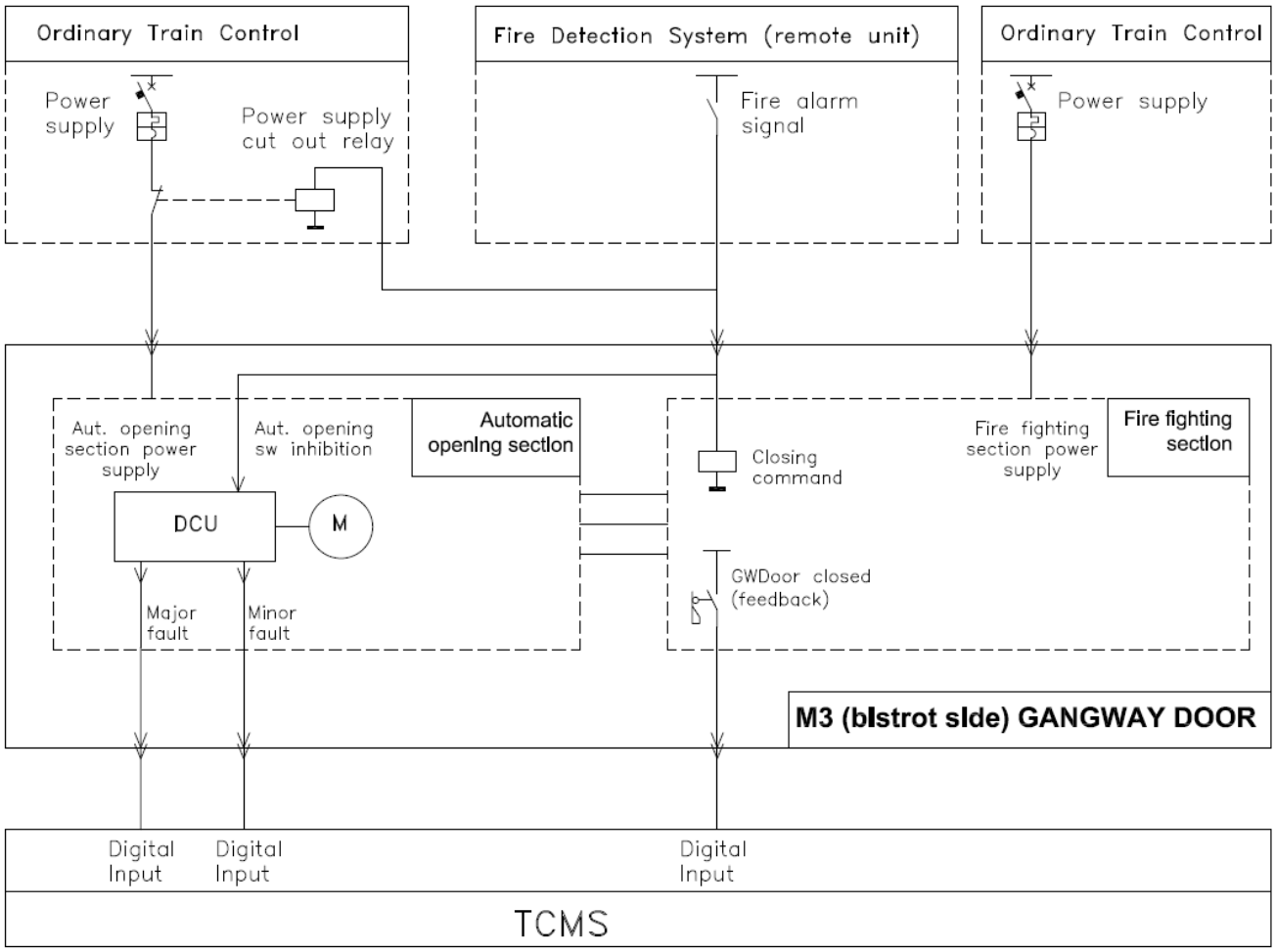



Figure 3 – M3 rear gangway door schematic

### 3.1.1.2.2 Normal condition

In normal condition the function is redundantly realized by means of HW components and DCU SW.

### 3.1.1.2.3 Failure cases

a) Loss of function of the HW power supply cut off. Loss of a redundancy (the DCU SW still inhibits the door opening).
b) Loss of function of the DCU SW inhibition function. Loss of a redundancy (the HW power supply cut off prevents the door from opening).

### 3.1.1.2.4 Failure detection

a) Failure condition detected by the TCMS comparing the FDS command with the major fault signal of the gangway door.
b) Not detected.

### 3.1.1.2.5 Failure handling

a) Driver has to act according to operational manual.

#### 3.1.1.3 Requirements

| | |
|---|---|
| **ID** | 2F_08.04.Zefiro-Europe.TRS.48 |
| **Description** | Both gangway doors of both gangways of the car where a fire has been detected shall automatically close. |
| **Safety level** | SIL 1 |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.61 | The gangway door shall receive the closing command by means of a hardwired line. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 1 |
| 2F_08.04.Zefiro-Europe.CONCEPT.97 | The gangway door shall close when the closing command hardwired line is energized | Derived Requirement | 4S_08.06.02.-.Gangway Doors | | SIL 1 |
| 2F_08.04.Zefiro-Europe.CONCEPT.130 | Power supply shall be cut to the GWDoor_R_M3 DCU as soon as the command from the fire detection system is energized. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 1 |
| 2F_08.04.Zefiro-Europe.CONCEPT.131 | DCU SW shall inhibit any door movement as soon as the command from the fire detection system is energized. | Derived Requirement | 4S_08.06.02.-.Gangway Doors | | SIL 0 |

## 3.2 Galley Doors

### 3.2.1 Function design

The DCUs of the 2 saloon doors in DM1 coach are connected one to the other in order to realize a synchronization function.

### 3.2.2 Normal condition

Once one of the 2 door is commanded to open, by the presence sensors or the bush button, the DCU of such door will command the DCU of the other door to open.

### 3.2.3 Failure cases

a) Loss of the synchronization function. The door can still be open by the presence sensor or the push button.

### 3.2.4 Failure detection

a) Not detected.

### 3.2.5 Failure handling

NA

### 3.2.6 Requirements

| | |
|---|---|
| **ID** | 2F_08.04.Zefiro-Europe.TRS.84 |
| **Description** | The 2 galley doors shall be synchronized and operated in parallel by the activation of one of the presence sensors or by one of the pushbuttons. |
| **Safety level** | SIL 0 |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.145 | A suitable hardwired connection shall be foreseen in order to transfer the synchronization signals. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | N/A |
| 2F_08.04.Zefiro- | Once one of the 2 galley door is commanded to | Derived | 4S_08.06.-.Interior | | SIL 0 |

| | | | | | |
|---|---|---|---|---|---|
| Europe.CONCEPT.146 | open, by the presence sensors or the bush button, the DCU of such door shall command the DCU of the other galley door to open | Requirement | Doors | | |

## 3.3 Supervision

### 3.3.1 Diagnostic data

#### 3.3.1.1 Function Design

Saloon and gangway doors are connected to the TCMS for diagnostic purpose using hardwired lines (two lines for each saloon door and one line for each gangway door).

Each saloon door and the GWDoor_R_M3 provide information about its actual status sending to the TCMS the following signals:

- Major fault (door out of order). This signal is normally at high level (+Batt level means door OK)
- Minor fault (the door is still able to work but maintenance is needed). This signal is normally at low level (+Batt level means maintenance needed)

Each gangway door provides information about its actual status sending to the TCMS the following signal:

- Closed position. This signal is normally at low level (+Batt means door closed)

This information is compared by the TCMS with the relevant closing command coming from the FDS in order to identify unexpected scenarios like:

- gangway door unexpected closed (without FDS command)
- gangway closing command failed (the FDS commands the closing of the gangway door due to a fire and the gangway door does not close)

#### 3.3.1.2 Requirements

| | |
|---|---|
| **ID** | 2F_08.04.Zefiro-Europe.TRS.63 |
| **Description** | Diagnostic data exchange between TCMS and internal doors shall be done by means of hardwired lines. |
| **Safety level** | SIL 0 |

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.66 | A major failure on a saloon door (SDoor_y_zzz_MjFault) shall be transmitted to TCMS | Derived Requirement | 4S_08.06.-.Interior Doors | | SIL 0 |
| 2F_08.04.Zefiro-Europe.CONCEPT.99 | The major failure of each saloon door (SDoor_y_zzz_MjFault) shall be transmitted to TCMS by means of a dedicated wire. | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |
| 2F_08.04.Zefiro-Europe.CONCEPT.100 | TCMS shall store an event (Warning) in case of a major fault of a saloon door (SDoor_y_zzz_MjFault). | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_08.04.Zefiro-Europe.CONCEPT.104 | A minor failure on a saloon door (SDoor_y_zzz_MnFault) shall be transmitted to TCMS | Derived Requirement | 4S_08.06.-.Interior Doors | | SIL 0 |
| 2F_08.04.Zefiro-Europe.CONCEPT.105 | The minor failure of each saloon door (SDoor_y_zzz_MnFault) shall be transmitted to TCMS by means of a dedicated wire | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |
| 2F_08.04.Zefiro-Europe.CONCEPT.101 | TCMS shall store an event (Maintenance) in case of a minor fault of a saloon door (SDoor_y_zzz_MnFault). | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_08.04.Zefiro-Europe.CONCEPT.147 | A major failure on the GWDoor_R_M3 (GWDoor_R_M3 _MjFault) shall be transmitted to TCMS | Derived Requirement | 4S_08.06.02.-.Gangway Doors | | SIL 0 |
| 2F_08.04.Zefiro- | The major failure of the GWDoor_R_M3 | Derived | 4S_09.02.09.- | | SIL 0 |

| | | | | | |
|---|---|---|---|---|---|
| Europe.CONCEPT.148 | (GWDoor_R_M3 _MjFault) shall be transmitted to TCMS by means of a dedicated wire. | Requirement | .Conventional Train Control - Circuits | | |
| 2F_08.04.Zefiro-Europe.CONCEPT.149 | TCMS shall store an event (Warning) in case of a major fault of the GWDoor_R_M3 (GWDoor_R_M3 _MjFault). | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_08.04.Zefiro-Europe.CONCEPT.150 | A minor failure on the GWDoor_R_M3 (GWDoor_R_M3 _MnFault) shall be transmitted to TCMS | Derived Requirement | 4S_08.06.02.-.Gangway Doors | | SIL 0 |
| 2F_08.04.Zefiro-Europe.CONCEPT.151 | The minor failure on the GWDoor_R_M3 (GWDoor_R_M3 _MnFault) shall be transmitted to TCMS by means of a dedicated wire | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |
| 2F_08.04.Zefiro-Europe.CONCEPT.152 | TCMS shall store an event (Maintenance) in case of a minor fault of the GWDoor_R_M3 (GWDoor_R_M3 _MnFault). | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |

Information about the gangway door status (closed position) is identified and sent to the TCMS at diagnostic purpose.
In addition, all gangway door closing commands (from the FDS) are sent to the TCMS in order to compare them with the actual status of the gangway door and identify unexpected scenarios

| Nr | Description | Type | Derived to | User Interface | SIL |
|---|---|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.108 | The gangway door closed position (GDoor_R_DM1_Closed ) shall be identified and sent to the TCMS. | Derived Requirement | 4S_08.06.02.-.Gangway Doors | | N/A |
| 2F_08.04.Zefiro-Europe.CONCEPT.109 | The gangway door closed position (GDoor_R_DM1_Closed) shall sent to the TCMS by means of a hardwired line | Derived Requirement | 4S_09.02.09.-.Conventional Train Control - Circuits | | SIL 0 |
| 2F_08.04.Zefiro-Europe.CONCEPT.107 | TCMS shall store an event (Warning) in case of a gangway door closes without command from the FDS. This requirement is not applicable to GWDoor_R_M3. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_08.04.Zefiro-Europe.CONCEPT.110 | TCMS shall store an event (Alarm) if the gangway is not closed 10 TBD seconds after the closing command from the FDS. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |
| 2F_08.04.Zefiro-Europe.CONCEPT.153 | TCMS shall store an event (Maintenance) if the signal GWDoor_R_M3 _MjFault is at +Batt level while the command from the fire detection system is active for that door. | Derived Requirement | 4S_09.03.05.-.TCMS Software | EVENT | N/A |

# 4 Signal Interface

## 4.1 I/O Interface

### 4.1.1 Supplier system

Inputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.28 | GWDoor_y_ZZZ_ClsCmd Closing command from FDS | | 2F_08.04.Zefiro-Europe.CONCEPT.97 |

Outputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.29 | SDoor_y_zzz_MjFault (Saloon door Major Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.66 |
| 2F_08.04.Zefiro-Europe.CONCEPT.112 | Sdoor_y_zzz_MnFault (Saloon door Minor Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.104 |
| 2F_08.04.Zefiro-Europe.CONCEPT. | GWdoor_y_zzz_Closed (Gangway door closed position) | | 2F_08.04.Zefiro-Europe.CONCEPT.108 |

| | | | |
|---|---|---|---|
| 111 | | | |
| 2F_08.04.Zefiro-Europe.CONCEPT.154 | GWDoor_R_M3 _MjFault (Gangway door Major Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.147 |
| 2F_08.04.Zefiro-Europe.CONCEPT.155 | GWDoor_R_M3 _MnFault (Gangway door Minor Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.150 |

### 4.1.2 TCMS SW

Inputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.30 | SDoor_y_zzz_MjFault (Saloon door Major Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.100 |
| 2F_08.04.Zefiro-Europe.CONCEPT.114 | Sdoor_y_zzz_MnFault (Saloon door Minor Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.101 |
| 2F_08.04.Zefiro-Europe.CONCEPT.113 | GWdoor_y_zzz_Closed (Gangway door closed position) | | 2F_08.04.Zefiro-Europe.CONCEPT.109 |
| 2F_08.04.Zefiro-Europe.CONCEPT.156 | GWDoor_R_M3 _MjFault (Gangway door Major Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.149 |
| 2F_08.04.Zefiro-Europe.CONCEPT.157 | GWDoor_R_M3 _MnFault (Gangway door Minor Fault) | | 2F_08.04.Zefiro-Europe.CONCEPT.152 |

Outputs:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.31 | -- | | |

## 4.2 Bus Interface

Supplier to TCMS SW:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.32 | -- | | |

TCMS SW to Supplier:

| Nr | Functional name | Type | Linked to requirement |
|---|---|---|---|
| 2F_08.04.Zefiro-Europe.CONCEPT.33 | -- | | |

# 5 References

[1] EC01P006798B TRD Interior Doors System
[2] EC01P006740B TRD Gangway Door System